SIMPLY SUPPORTED STEEL BEAM CALCULATION

ULS bending and shear check of a simply supported beam (EN 1990 load factors).

1. DESIGN PARAMETERS

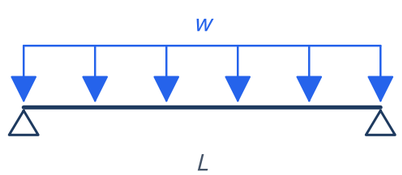
Span, tributary width and characteristic area loads.

$L := 8.5 \cdot \mathrm{m}$ $b := 2 \cdot \mathrm{m}$

$G := 5 \cdot \mathrm{kPa}$ $Q := 5 \cdot \mathrm{kPa}$

2. DESIGN LOADING

$$w := (1.35G + 1.5Q)\, b = 28.5 \cdot \mathrm{kN/m}$$



3. BENDING MOMENT

$$\begin{aligned} M_{Ed} &:= \frac{wL^2}{8} \\ &= \frac{(28.5 \cdot \mathrm{kN/m}) \cdot (8.5 \cdot \mathrm{m})^2}{8} \\ &= 257.39 \cdot \mathrm{kN \cdot m} \end{aligned}$$

4. SHEAR FORCE

$$V_{Ed} := \frac{wL}{2} = 121.13 \cdot \mathrm{kN}$$

5. CAPACITY CHECK

$M_{Rd} := 300 \cdot \mathrm{kN \cdot m}$

$$\eta_M := \frac{M_{Ed}}{M_{Rd}} = 0.858$$

$\eta_M = 0.858 \leq 1$ PASS